

Flatrim Dreshaj

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Education

Bachelor of Science in Electrical Engineering
GPA: 3.55

Expected graduation date: May 2027

Iowa State University

Skills : Keysight ADS, KiCAD, Microwave CST, Python, MATLAB

Relevant Classes: Microwave Engineering, High Speed Systems Design, Communication Systems, Wave Propagation and TLines

Work Experience:

Applied Microwave Testing Lab Researcher

May 2025- Current

- Designed and programmed a Synthetic Aperture Radar scanner program in Python that utilized VNAs and motor controllers for 0 - 240 GHz Microwave imaging.
- Wrote Python scripts for controlling VNAs and custom radar hardware designs through PyVISA, Socket and FTDI, gathering S parameter data and transferring to an HDF5 file from Anritsu and Keysight VNAs.
- Wrote Python scripts for controlling motor drivers GM215, Octopus Pro, to move point to point in a raster scan pattern, stop at each point and send a command to the VNA to take a measurement at that point.
- Wrote a threading system in Python which transferred and wrote VNA data while the motor was moving, reducing total scan time compared to the previous LabVIEW program.
- Built 3D Microwave Imaging system for doing arbitrary curvature structure imaging.
- Build a novel 4 axis microwave imaging system for ongoing research efforts of SVD imaging.
- Built a 2m long Microwave Imaging scanner for imaging larger area structures like concrete blocks.
- Developed SAR Algorithms to turn gathered S Parameter data from imaging systems into 3d images at each depth slice of a material.
- Designed and tested 9-20 GHz Vivaldi Antennas using Anritsu VNAs.
- Designed 1-100 GHz Antipodal Vivaldi Antennas.
- Designed and Simulated Ka, Ku, S band and X band waveguides, SMA V and K Connectors, Magnetrons, Coax - Waveguide transitions, Microstrip to slotline transitions, Traveling Wave Tube Amplifiers.
- System building Research FMCW Radar using ADAR2001, 2004 eval boards with a FM Anritsu Signal Generator, used Signal Hound spectrum analyzers to measure the system, took differential IF out lines of ADAR2004 through an ethernet cable and used Picoscope oscilloscopes to analyze the time domain response.
- Designed Nicolson-Ross-Weir(NRW) method for extracting complex permittivity and permeability from calibrated S parameter measurements.

Projects :

VNA PCB Design

Aug 2025 - Current

- Designed 6 Layer, 2 port VNA at a 0.5 - 10GHz range with a goal of a >130dB dynamic range and oscillator phase noise of less than -110dBc/Hz at 1MHz offset.
- Design and Layout of a STM32H7 Based VNA motherboard that will take the output of the downconverters and sample them using a high speed dual ADC.
- Motherboard design including 100 MBps Ethernet for S parameter data transfer, USB-C Full Speed, and SWD for debugging.
- Design and Layout in Keysight ADS of a 2.3 - 2.8GHz Common Base Colpitts VCO in Keysight ADS providing a power output of -1.8 dBm to -11dBm to a 50 Ohm load over the tuning capacitance of 0.7-7.8pF. Phase Noise simulations of -169.1, -173.1, -171.3, -169.9, -169.6, 167.7 and -165.1 dBc at a 1 MHz offset for capacitances of 1-7 pF respectively.
- Design and Layout in ADS of a 90 degree hybrid directional coupler for 2.45GHz providing -3dB coupling and -44dB isolation to be used for separating VCO DUT input and reference signals, along with additional utility in the downconverter design.
- Design of a 180 degree Modified Rat race coupler providing -3dB coupling over a 2 GHz bandwidth.
- Design of a balanced hybrid mixer providing -5.6dB at IF.
- Design and Layout in ADS of a 2.45 GHz amplifier, providing an S11 of -34.3dB, S22 of -17.57, S21 gain of 15.47dB from a maximum stable gain of 16.1dB.
- 2.45 GHz 6% Fractional bandwidth Gap Coupled Bandpass Filter Design and Layout getting a -25.7 dB S11 and -5.33dB S21 in the passband.
- Designed a fast sweep DAC board in KiCad around the AD3551R with an external transimpedance amplifier to drive the VCO tuning voltage for 1-10GHz output waveforms.
- Designed SOLT and TRL calibration algorithm software, to be used after processing of S parameters.
- Designed 1-10GHz VCO Board.
- Issues to be solved are wideband coupler design and wideband mixer design although I am planning on just making a board using ICs that solve these issues.